END-SESSION CAPSTONE · 30 MINUTES

Design an agent on paper

A fresh business ask. Eight PRD questions. One HLD sketch. One failure pre-mortem. ChatGPT is the partner; the deliverable is a one-page design doc you can show in a Monday standup.

THE BUSINESS ASK

> *"We want an agent to help our finance team chase overdue invoices. It should look up which customers are past due, check whether a reminder was already sent this week, draft a personalised reminder email, and schedule the send. Spends over $0 in third-party tools (e.g., paid lookups) need finance manager approval."*

1 Classify first (3 min)

Before designing anything, run the four-quadrant rubric on the business ask. Open ChatGPT and paste:

PROMPT 1

```
I am designing an AI system. The business ask is:

"An agent that chases overdue invoices: look up past-due customers, check if a
reminder went out this week, draft a personalised reminder email, and schedule
the send. Spends > $0 on paid lookups need finance manager approval."

Apply the four-quadrant rubric:
- Code-decided vs model-decided next step?
- Single step or multi-step?

Classify into AUTOMATION / WORKFLOW / LLM CALL / AGENT. Justify in two sentences.
Be honest — if it is actually a workflow, say so.
```

2 Build the PRD with eight answers (10 min)

Whatever the classification, draft a PRD using the eight questions. ChatGPT helps; you decide. Paste the next prompt:

PROMPT 2

```
Now produce a one-page PRD for this system. Answer all eight questions in a
numbered list. One sentence per answer.

1. Who is the user?
2. What is the goal? (one sentence)
3. Success KPI — measurable, not "better experience"
4. Autonomy boundary — what is free, what needs approval, what is prohibited
5. What does failure look like? (concrete)
6. Rollback path — what reverses on partial failure?
7. PII surface — what data is touched, how is it masked
8. Cost ceiling — per session, per day

Rules:
- Be specific. "Better experience" and "as needed" are banned.
- The autonomy boundary must name actual operations.
- The KPI must be measurable with a number and a time window.
```

3 Draft the HLD as Mermaid (8 min)

Now ask ChatGPT to produce a Mermaid flowchart for the HLD. This is what would go in the repo's README.

PROMPT 3

```
Based on the PRD you just wrote, produce a Mermaid flowchart for the HLD. Use
`flowchart TD` syntax. Include:
- The user / trigger as the top node
- The planner / orchestrator
- Any sub-agents (only if the design genuinely needs them)
- The tools each component calls (name them: customer_lookup, reminder_history,
draft_email, schedule_send, etc.)
- The approval gate (HITL) before any action that crosses the autonomy boundary
- Memory nodes if relevant

Rules:
- One Mermaid block. Should fit in a single screen.
- Every tool you name must trace to one of the PRD's operations.
- If you add a sub-agent, justify it in one line below the diagram.
```

VERIFY MERMAID RENDERS

Copy the code block ChatGPT produces. Paste it into **mermaid.live** (free, no signup). If it renders, your diagram is valid. If it errors, ask ChatGPT to fix the syntax and re-paste.

4 Failure pre-mortem (7 min)

Now stress-test your design against the six failure patterns you learned today.

PROMPT 4

```
Pre-mortem the design. For each of the six failure patterns (looping, silent tool
failure, hallucinated step, context poisoning, state drift, unsafe autonomy),
produce one table row:

- Pattern name
- Where it would bite THIS invoice-chasing system specifically
- One concrete defence implementable in the HLD

Then highlight the TOP TWO highest-risk patterns for this system and explain in
one line each why those are the most likely to actually happen in production.
```

5 Self-verify the design (2 min)

Before you submit, run this checklist. If any box stays unchecked, fix it.

- ☐ The PRD has all eight answers, each one sentence, none of them "as needed" or "better experience."
- ☐ The autonomy boundary names actual operations (not "sensitive actions").
- ☐ The Success KPI has a number and a time window (e.g., "60% within 5 days").
- ☐ The Mermaid renders cleanly in mermaid.live.
- ☐ Every tool in the diagram traces to an operation in the PRD.
- ☐ The approval gate appears in the diagram before any irreversible action.
- ☐ Every failure pattern has a domain-specific risk (no generic "add error handling").
- ☐ The top-two highest-risk patterns are identified and justified.

DELIVERABLE

One ChatGPT conversation with four artefacts: classification, PRD, Mermaid HLD, failure pre-mortem. Copy the whole thing into a doc or screenshot it. **This is exactly the format your Day 2 build session expects.**